

# Corporate Finance — Exam Cheat Sheet

FIN211 · Every formula, decision rule and key definition from Lectures 1–13, with the course's worked examples (tap “Example” to expand).

Sources: Titman & Keown, Financial Management (13e) · Gitman, Principles of Managerial Finance (15e)

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## 1 · Foundations & Key Terms

Lec 1 · Lec 2-P1

### Finance

Management of money by individuals (personal), corporations (corporate), governments (public).

### Corporate finance

Deals with capital structure (debt + equity); maximizes shareholders' wealth via short- & long-term financing plans.

### Corporate goal

Maximize shareholder wealth (share price), not merely profit.

### Capital management

Provide short-term liquidity through debt & equity. Too much debt → default risk; too much equity → lower EPS.

### Investment

Purchase of assets (stocks, bonds, real estate...) to earn dividends, interest, capital gains.

### Capital budgeting

Decision process for analyzing potential investments in fixed assets.

### Budgeting

Planning money by estimating future income/expenses over a period.

### Financial markets

Any marketplace where securities trade (stock, bond, forex, derivatives).

### Financial services

Design & delivery of advice and financial products to individuals, firms, governments.

### Areas of corporate finance

Capital budgeting · capital financing · working-capital management.

## FIVE PRINCIPLES OF FINANCE

- **Time value of money** — a dollar today > a dollar tomorrow.
- **Risk-return trade-off** — higher risk ⇒ higher return.
- **Cash flows are the source of value** (cash that can be taken out of the business).
- **Market prices reflect information** (buy & sell ⇒ equilibrium).
- **Individuals respond to incentives** (motivation).

## BUSINESS FORMS

|                                | SOLE PROPRIETORSHIP | PARTNERSHIP      | LP / LLC                 | CORPORATION                          |
|--------------------------------|---------------------|------------------|--------------------------|--------------------------------------|
| Owners                         | One                 | Two or more      | ≥1 GP + limited LPs      | Shareholders                         |
| Liability for debts            | Unlimited           | Unlimited (each) | GP unlimited, LP limited | Limited                              |
| Owners manage?                 | Yes                 | Yes              | GP yes, LP no            | No (managers)                        |
| Change of ownership dissolves? | Yes                 | Yes              | GP yes, LP no            | No                                   |
| Access to capital              | Limited             | Limited          | Size-dependent           | Very easy                            |
| Taxation                       | Personal            | Personal         | Personal                 | <b>Double</b> (earnings + dividends) |

## DEBT VS EQUITY

|                        | DEBT                       | EQUITY       |
|------------------------|----------------------------|--------------|
| Voting rights          | No                         | Yes          |
| Claim on income/assets | Senior                     | Subordinate  |
| Maturity               | Stated                     | None         |
| Tax treatment          | Interest <b>deductible</b> | No deduction |

## ACCOUNTING VS FINANCE

- Focus: accounting = past & present · finance = **future-oriented** decisions.
- Basis: accounting = **accrual** (IFRS/GAAP) · finance = **cash** (risk, TVM, value).

## EARNINGS PER SHARE

EPS =

Net income – Preferred dividends

Outstanding common shares

### Example — EPS

NI 1,000,000; pref. div. 100,000; 450,000 shares → EPS = 900,000/450,000 = **\$2.00**. Bakery: (20,000–2,000)/10,000 = **2.0**. More shares issued ⇒ lower EPS.

### CASH-FLOW CATEGORIES

- **Operating:** cash sales, collecting A/R, tax refunds | paying wages, payables, overhead, interest.
- **Investing:** sale / purchase of fixed assets.
- **Financing:** sale of stock, borrowing | buying back stock, dividends, debt repayment.

### DEPRECIATION (MACRS, 5-YEAR RECOVERY)

Rates: Yr1 **20%** · Yr2 **32%** · Yr3 **19%** · Yr4 **12%** · Yr5 **12%** · Yr6 **5%** →  $Dep_t = \text{Installed cost} \times \text{rate}_t$

#### Example — \$100,000 machine

Depreciation:

20,000 · 32,000 · 19,000 · 12,000 · 12,000 · 5,000 (total 100,000). Depreciation matters: non-cash charge + **tax shield**.

### OPERATING CASH FLOW

$NOPAT = EBIT \times (1 - T)$  | **OCF (unlevered)** =  $EBIT \times (1 - T) + \text{Depreciation}$

**OCF (levered)** = Net income after tax + Depreciation

Unlevered = cash to all capital providers before debt & equity; levered = after interest/debt servicing.

#### Example — Baker Corp.

$OCF = 370 \times (1 - .40) + 100 = \text{\$322}$  (thousands).

### FREE CASH FLOW (FCF)

**FCF** =  $OCF - NFAI - NCAI$

$NFAI = \Delta \text{ Net fixed assets} + \text{Depreciation}$  |  $NCAI = \Delta \text{ Current assets} - \Delta (A/P + \text{Accruals})$

#### Example — Baker Corp.

$NFAI = (1,200 - 1,000) + 100 = 300$  ·  $NCAI = (2,000 - 1,900) - (800 - 700) = 0$  ·  $FCF = 322 - 300 - 0 = \text{\$22}$ .

### FREE CASH FLOW TO THE FIRM (FCFF) ★

**FCFF** =  $NI + \text{Depreciation} + [\text{Interest} \times (1 - T)] - \text{Fixed-asset investment} - \Delta \text{ Working capital}$

FCFF = cash available to debt holders, equity holders and re-investment.

#### Example 1

NI 500, D&A 120, Int 100, T 30%, FA inv 200, WC inv 50 →  $FCFF = 500 + 120 + 100(0.70) - 200 - 50 = \text{\$440}$ .

#### Example 2 — ABC Co.

$NI 245,000 + \text{Dep } 100,000 + \text{Int } 50,000(0.70) - \text{FA inv } (850,000 - 600,000) - \Delta WC [(440,000 - 180,000) - (350,000 - 150,000)] = 245,000 + 100,000 + 35,000 - 250,000 - 60,000 = \text{\$70,000}$ .

CASH-FLOW RATIOS

Cash-flow margin =  $\frac{\text{OCF}}{\text{Sales revenue}}$  | Cash-flow coverage =  $\frac{\text{OCF}}{\text{Total debt}}$

Margin: high = strong cash conversion; negative = operations consume cash. Coverage: high = debt repaid quickly ( $>1 \Rightarrow$  under one year); 0.1-0.3 acceptable;  $<0.1$  low; negative = financial risk. ABC: margin =  $345,000/2,000,000 = 0.1725$ ; coverage =  $345,000/450,000 \approx 0.77$ .

### 3 · Cash Budgeting

Lec 4-P1

Financial planning = **cash planning** (cash budget) + **profit planning** (pro-forma statements). The cash budget shows expected cash **shortages** / **surpluses** month by month.

#### THE THREE STEPS

- **Step 1 — Cash receipts:** cash sales (% of sales) + collections of A/R lagged 1 month and 2 months + other receipts.
- **Step 2 — Cash disbursements:** purchases (% of sales) paid cash / lagged 1 mo / lagged 2 mo + rent, wages & salaries, taxes, fixed-asset outlays, interest, dividends, principal.
- **Step 3 — Financing position:**

Net cash flow = Receipts – Disbursements → Ending cash = Net CF + Beginning cash

Ending < Minimum = **Required total financing** (notes payable) | Ending > Minimum = **Excess cash** (marketable securities)

**Example — Halley Co. (Oct-Dec, \$000; sales Aug 100, Sep 200, Oct 400, Nov 300, Dec 200; 20% cash, 50% +1mo, 30% +2mo; Dec dividend +30)**

|                                                                              | OCT | NOV  | DEC  |
|------------------------------------------------------------------------------|-----|------|------|
| Total cash receipts                                                          | 210 | 320  | 340  |
| Less: disbursements (purchases = 70% of sales; 10% cash, 70% +1mo, 20% +2mo) | 213 | 418  | 305  |
| = Net cash flow                                                              | (3) | (98) | 35   |
| + Beginning cash                                                             | 50  | 47   | (51) |
| = Ending cash                                                                | 47  | (51) | (16) |
| – Minimum cash balance                                                       | 25  | 25   | 25   |
| Required financing (notes payable)                                           | —   | 76   | 41   |
| Excess cash (marketable securities)                                          | 22  | —    | —    |

Oct surplus 22 → invest in marketable securities; Nov & Dec deficits → finance.

**Example — Q1 2025 (beginning cash 50, minimum 25)**

|                             | JAN       | FEB      | MAR      |
|-----------------------------|-----------|----------|----------|
| Receipts – Disbursements    | 420–426   | 640–836  | 680–610  |
| = Net cash flow             | (6)       | (196)    | 70       |
| + Beginning cash            | 50        | 44       | (152)    |
| = Ending cash               | 44        | (152)    | (82)     |
| Required financing / Excess | excess 19 | fin. 177 | fin. 107 |

## COMMON-SIZE ANALYSIS

$$\text{Common-size \%} = \frac{\text{Comparison amount}}{\text{Base amount}} \times 100$$

Base = total revenues (income statement) or total assets (balance sheet). E.g. COGS 98,432/218,739 = 45%; Equity/TA 344,373/365,389 = 94.2%.

## TREND (HORIZONTAL) ANALYSIS

$$\% \text{ Change} = \frac{\text{Dollar change}}{\text{Base-year amount}} \times 100, \quad \text{Dollar change} = \text{Analysis year} - \text{Base year}$$

E.g. Cash (261,248–224,399)/224,399 = 16.4%; NI (25–15)/15 = 66.6%; RE (67–65)/65 = 3.07%.

## RATIO TOOLKIT ★

| GROUP         | RATIO                 | FORMULA                            | READ / RULE                               |
|---------------|-----------------------|------------------------------------|-------------------------------------------|
| Liquidity     | Current ratio         | CA / CL                            | ≥ 1 → covers short-term liabilities       |
|               | Quick (acid-test)     | (Cash + ST investments + A/R) / CL | ≥ 1 → covers CL without selling inventory |
|               | Cash ratio            | Cash (& equivalents) / CL          | Higher = stronger position                |
| Efficiency    | A/R turnover          | Net credit sales / Avg net A/R     | Higher = collects debts efficiently       |
|               | Inventory turnover    | COGS / Avg inventory               | Higher = efficient inventory management   |
| Solvency      | Debt to assets        | Total debt / Total assets          | Long-term survival; lower = safer         |
|               | Times interest earned | EBIT / Interest expense            | Ability to meet interest payments         |
| Profitability | Profit margin         | Net income / Net sales             | Net income per \$1 of sales               |
|               | Asset turnover        | Net sales / Avg total assets       | Revenue per \$1 of assets                 |
|               | Return on assets      | Net income / Avg total assets      | NI per \$1 invested in assets             |
|               | Return on equity      | Net income / Avg equity            | NI per \$1 of equity                      |
|               | EPS                   | NI / Weighted-avg shares           | —                                         |
|               | P/E ratio             | Market price per share / EPS       | —                                         |

### Example — ABC Co. (all ratios)

Current = 440/180 = **2.44** · Quick = (100+180)/180 = **1.56** · Cash = 100/180 = **0.56** · Debt/Assets = (180+450)/1,290 = **48.8%** · TIE = 400/50 = **8x** · Margin = 245/2,000 = **12.25%** · Asset turnover = 2,000/[(950+1,290)/2] = **1.79** · ROA = 245/1,120 = **21.87%** · ROE = 245/[(500+660)/2] = **42.24%** · EPS = 245,000/250,000 = **\$0.98** · P/E = 5/0.98 = **5.1**.

## EXPECTED RETURN (DIVIDENDS + CAPITAL GAINS)

$$E(r) = \frac{D}{P} + g$$

D = dividend, P = price, g = growth. E.g. \$1 dividend on \$25 stock, g = 7% →  $1/25 + 0.07 = 11\%$ .

## ZERO-GROWTH MODEL (PERPETUITY; ALSO PREFERRED STOCK)

$$P_0 = D_1 \times \frac{1}{r_s} = \frac{D_1}{r_s}$$

### Examples

Constant \$3 dividend,  $r_s = 15\% \rightarrow P_0 = 3/0.15 = \mathbf{\$20}$ . Table:  $3/0.12 = 25 \cdot 5/0.10 = 50 \cdot 4/0.16 = 25 \cdot 2.5/0.08 = 31.25 \cdot 6/0.15 = 40$ .

## CONSTANT-GROWTH MODEL (GORDON) ★

$$P_0 = \frac{D_1}{r_s - g}, \quad D_1 = D_0(1 + g)$$

### Examples

$D_0 = 1.40$ ,  $g = 7\%$ ,  $r_s = 15\% \rightarrow 1.40(1.07)/(0.15 - 0.07) = \mathbf{\$18.75}$ . Table:  $2/(0.10 - 0.04) = 33.33 \cdot 3/(0.12 - 0.05) = 42.86 \cdot 1.5/(0.09 - 0.03) = 25 \cdot 2.5/(0.11 - 0.06) = 50 \cdot 4/(0.14 - 0.05) = 44.44$ .

Buy if value > price (undervalued)

Sell/avoid if value < price (overvalued)

## MUST-KNOW TERMS

- **Common stock:** residual claimants; voting rights; returns = dividends + capital gains.
- **Share types:** authorized · issued · outstanding · treasury; par vs no-par.
- **IPO:** needs shareholder approval + investment bank underwriting (bank buys issue at agreed price, bears selling risk).
- **Preferred stock:** fixed dividend, prior claim, **cumulative** (arrears), paid before common; hybrid of bond & stock.

## INDIVIDUAL ASSET

$$E(R) = \sum P_i R_i \quad | \quad \sigma^2 = \sum P_i (R_i - E(R))^2 \quad | \quad \sigma = \sqrt{\sigma^2}$$

### Example — states of nature (P: recession .2, normal .5, boom .3)

Stock C:  $E(R) = .2(.02) + .5(.10) + .3(.15) = 9.7\%$ ;  $\sigma^2 = 0.002028 \rightarrow \sigma = 4.5\%$ . Stock T:  $E(R) = 17.7\%$ . Total risk =  $\sigma$ ; systematic risk =  $\beta$  (C:  $\sigma 20\% / \beta 1.25$  vs K:  $\sigma 30\% / \beta 0.95 \rightarrow$  K more total risk, C more systematic risk & higher expected return).

## PORTFOLIO RETURN & WEIGHTS

$$w_i = \frac{\text{Investment}_i}{\text{Total portfolio}} \quad (\sum w = 1) \quad | \quad E(R_p) = \sum w_i E(R_i)$$

### Examples

\$15,000: IREX 2,000 (.1333, 19.65%) · AIND 3,000 (.2, 8.96%) · EFID 4,000 (.266, 9.67%) · EKHO 6,000 (.4, 8.13%)  $\rightarrow E(R_p) = .0262 + .0179 + .0257 + .0325 = 10\%$ . AbuAuf & VCola — half the money in each, 14% return each  $\rightarrow 14\%$ .

## TWO-ASSET PORTFOLIO STANDARD DEVIATION ★

$$\sigma_p = \sqrt{(w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2)}$$

### Example — w .5/.5, $\sigma$ .2/.2, $\rho = +0.75$

$\sigma_p = \sqrt{(.5^2 \cdot .2^2 + .5^2 \cdot .2^2 + 2 \cdot .5 \cdot .5 \cdot .75 \cdot .2 \cdot .2)} = \sqrt{0.035} = 0.187$ .  $\rho = +0.75 \Rightarrow$  returns move together but not in perfect unison.

## TYPES OF RISK

- **Systematic** (market / non-diversifiable): GDP, inflation, interest rates — measured by  $\beta$ .
- **Unsystematic** (unique / asset-specific): strikes, part shortages — **eliminated by diversification**.
- **Total risk** =  $\sigma$ ; don't hold one asset or one industry.

## PORTFOLIO BETA

$$\beta_p = \sum w_i \beta_i$$

$$.133(3.69) + .2(0.64) + .167(1.64) + .4(1.79) = 1.61.$$

## CAPM ★

$$R_e = R_f + \beta (R_m - R_f)$$

$R_f$  = risk-free rate (e.g. T-bills),  $R_m$  = market return,  $(R_m - R_f)$  = market risk premium. Higher  $\beta \Rightarrow$  higher risk premium.

### Examples & decision rule

$R_f$  4%,  $R_m$  10%,  $\beta$  1.5  $\rightarrow R_e = 4 + 1.5(6) = 13\%$ .  $R_f$  3%,  $R_m$  11% (premium 8%): A  $\beta .8 \rightarrow 9.4\%$  vs exp 8%  $\Rightarrow$  **No**; B  $\beta 1.2 \rightarrow 12.6\%$  vs 14%  $\Rightarrow$  **Yes**; C  $\beta 1.5 \rightarrow 15\%$  vs 16%  $\Rightarrow$  **Yes**.  $R_f$  5%,  $R_m$  12%: X  $\beta .5 \rightarrow 8.5\%$  vs 7% **No**; Y  $\beta 1.0 \rightarrow 12\%$  vs 12% **Yes**; Z  $\beta 1.8 \rightarrow 17.6\%$  vs 18% **Yes**.

**Accept: Expected return  $\geq$  CAPM return**

**Reject: Expected return  $<$  CAPM return**



## SIMPLE INTEREST

$$FV = PV \times (1 + r \cdot n) \quad | \quad PV = \frac{FV}{1 + r \cdot n}$$

PV 100 at 6% for 1 year → FV 106. FV table (PV, r, n → FV): (10,000, 5%, 3) → 11,500 · (25,000, 7%, 4) → 32,000 · (5,000, 6%, 1.5) → 5,450. PV table (FV, r, n → PV): (12,000, 5%, 3) → 10,434.78 · (20,000, 6%, 4) → 16,129.03 · (8,000, 9%, 2) → 6,779.66.

## COMPOUND INTEREST

$$FV = PV \times (1 + r)^n \quad | \quad PV = \frac{FV}{(1 + r)^n}$$

PV 100 at 6% for 5 years → FV 133.8. FV table (PV, r, n → FV): (10,000, 5%, 3) → 11,576.25 · (15,000, 8%, 5) → 22,039.92 · (50,000, 6%, 10) → 89,542.4. PV table (FV, r, n → PV): (15,000, 5%, 4) → 12,340 · (30,000, 6%, 3) → 25,188 · (10,000, 8%, 10) → 4,631.93.

## ANNUITIES (END-OF-YEAR PAYMENTS)

$$FV = PMT \times \frac{(1 + r)^n - 1}{r} \quad | \quad PV = PMT \times \frac{1 - (1 + r)^{-n}}{r}$$

PMT = 5,000 paid at the end of each year for 5 years, at r = 5%: FV = 5,000 × 5.5256 = **27,628**; PV = 5,000 × 4.3298 = **21,647.8**.

## BOND TERMS

- **Loans** = private debt (two parties). Pros: speed · reduced costs · financing flexibility. Cons: interest cost higher than bonds · restrictive covenants · possible future SEC registration. **Bonds** = public debt, traded in bond markets.
- **Indenture** = legal contract with trustee; bondholders' claims **senior** to common stock.
- **Par value** = face repaid at **maturity**; **coupon rate** = % of par paid yearly.
- **Call provision** = issuer may redeem early; **conversion** = bond → shares.
- **Floating rate** = index (LIBOR) of prior period + spread, subject to cap & floor.

$$\text{Current yield} = \frac{\text{Annual interest}}{\text{Current market price}} \quad | \quad \text{Accrued interest} = \frac{\text{Months since payment}}{12} \times \text{Annual coupon}$$

8% coupon, \$1,000 par, price \$700 → 80/700 = **11.4%**; 6 months accrued = 6/12 × 80 = **\$40**. Floating loan (LIBOR+0.25, cap 2.5, floor 1.75): wk2 2.23 · wk3 1.91 · wk4 1.77 · wk5 1.60 · wk6 1.85 · wk7 1.88 · wk8 1.92 · wk9 2.13 · wk10 2.18.

## BOND VALUE ★

$$B_0 = C \times \frac{1 - (1 + r)^{-n}}{r} + \frac{\text{Par}}{(1 + r)^n}$$

= PV of coupon annuity + PV of par. C = coupon rate × par.

### Examples

12% required, \$120 coupon, 5yr  $\rightarrow 432.576 + 567.427 = \mathbf{1,000}$  (par). 15%  $\rightarrow 402.264 + 497.77 = \mathbf{899.44}$  (discount). 9%  $\rightarrow 466.758 + 649.93 = \mathbf{1,116.68}$  (premium). Coupon 8% with required yield 11.4% over 10 years  $\rightarrow 458 + 339 = \mathbf{803}$ .  
Table (coupon rate, required yield  $r$ , years  $\rightarrow$  bond value): (7%, 5%, 3)  $\rightarrow 1,054.47$  · (8%, 10%, 4)  $\rightarrow 936.60$  · (6%, 7%, 5)  $\rightarrow 959.00$ .

$r > \text{coupon} \Rightarrow \text{discount}$  ·  $r = \text{coupon} \Rightarrow \text{par}$  ·  $r < \text{coupon} \Rightarrow \text{premium}$ .

**Accept/buy: DCF value > market price (undervalued)**

**Reject: DCF value < market price (overvalued)**

## 8 · Capital Budgeting

Lec 12

**Capital budgeting** = decision process for analyzing potential fixed-asset investments. Important because of: growth, large amounts, irreversibility, complexity, risk, long-term implications. Uses: new product/service, replacement, expansion, R&D, long-term contracts, safety/environment.

### TECHNIQUES

- **Non-discounted:** Payback period (PBP) · Average rate of return (ARR).
- **Discounted:** Discounted PBP · NPV · IRR · Profitability index.

### PAYBACK PERIOD

$$\text{Constant inflow: PBP} = \frac{\text{Initial investment}}{\text{Annual fixed cash inflow}}$$

$$\text{Variable inflow: PBP} = \text{Completed years} + \frac{\text{Remaining outflow}}{\text{Cash inflow of year } t}$$

#### Examples

Constant: A  $50,000/12,000 = 4.17\text{yr}$  · B  $60,000/15,000 = \mathbf{4\text{yr (best)}}$  · C  $45,000/10,000 = 4.5\text{yr}$ .

Variable: A  $50,000 \rightarrow \text{cum } 15/30/40/46 + 4,000/5,000 = \mathbf{4.8\text{yr}}$ ; B  $60,000 \rightarrow \text{cum } 20/40/50/56 + 4,000/6,000 = \mathbf{4.66\text{yr (best)}}$ ; C  $45,000 \rightarrow \text{cum } 12/24/36/42 + 3,000/4,000 = \mathbf{4.75\text{yr}}$ . **Shorter PBP = better**

### AVERAGE RATE OF RETURN

$$\text{ARR} = \frac{\text{Average profit after tax}}{\text{Average investment}} \quad \text{or} \quad \frac{\text{Average profit after tax}}{\text{Initial investment}}$$

Determines the average annual earning of a project.

- **MNC:** produces & sells in more than one country; parent in home country + typically 5–6 foreign subsidiaries with strategic interaction.
- **Classifications:** international · multinational · transnational corporations.
- **Formation (FDI):** buying an entire company, or partial ownership ( $\geq 10\%$  of stock).
- **Motives:** raw-material seekers (BP, Exxon, Total) · market seekers (per-capita markets) · cost minimizers (Microsoft).
- **Goal:** maximize shareholder wealth (share price + coupons).
- **Roles:** global manager (adjust to volatility) · politician & labor-union face (jobs/exports at home vs abroad).
- **Difficulties:** exchange-rate risk · inflation risk · tax-rate differences · multiple financial markets · political risk.
- **FX market participants:** bank & non-bank dealers · individuals & firms · speculators & arbitragers · central banks & treasuries · brokers.

### EXCHANGE-RATE REGIMES

- **Hard pegs:** gave up monetary sovereignty (adopt another currency / currency board).
- **Soft pegs:** fixed exchange rates.
- **Floating:** freely floating (market supply & demand, occasional intervention) · managed floating (secret targets, heavy intervention).
- **Residual:** everything else.

### QUOTATIONS

Direct = home-currency price of 1 unit foreign | Indirect = foreign-currency price of 1 unit home

European terms = indirect for Americans; American terms = direct. E.g. 22-Jan-2020: \$1 = EGP 15.80 = 0.90 € = 2.82 TD.

# ★ Decision-Rule Quick Card

all lectures

| SITUATION                                | RULE                                                                                                                                          |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Asset vs CAPM (Lec 9)                    | <div>Accept if <math>E(R) \geq R_f + \beta(R_m - R_f)</math></div> <div>Reject if below</div>                                                 |
| Stock / bond vs market price (Lec 8, 10) | <div>Buy if model value &gt; price (undervalued)</div> <div>Avoid if value &lt; price (overvalued)</div>                                      |
| Bond coupon vs required yield (Lec 10)   | $r > \text{coupon} \rightarrow \text{discount}$ · $r = \text{coupon} \rightarrow \text{par}$ · $r < \text{coupon} \rightarrow \text{premium}$ |
| Projects by payback (Lec 12)             | Choose shortest payback period                                                                                                                |
| Cash budget ending cash (Lec 4)          | Below minimum → notes payable (finance) · above minimum → marketable securities (invest)                                                      |
| Liquidity ratios (Lec 5)                 | Current & quick $\geq 1$ = able to cover short-term obligations                                                                               |
| Cash-flow coverage (Lec 3)               | $>1 \Rightarrow$ debt repayable in $<1$ yr · 0.1–0.3 acceptable · $<0.1$ low · negative = financial risk                                      |
| Diversification (Lec 9)                  | Only <b>unsystematic</b> risk can be diversified away; systematic ( $\beta$ ) cannot                                                          |

Course assessment: classwork 20% · mid-term I 20% · mid-term II 20% · final 40%. Good luck! ☐